Final Report:

Study Overview:

The primary goal of this study is to understand if there is a connection between re-election incentives and aversion to corruption for Brazilian mayors. This causal question can be studied by examining an observational dataset that will, through double/debiased machine learning techniques (Chernozhukov et al. 2018), allow for the discovery of salient evidence in support of the hypothesis. With that said, it is necessary to acknowledge that such a study is incapable of observing intangible aspects of a mayor such as charisma, which, in reality, could allow certain mayors to engage in corruption without penalty.

For context on the scenario surrounding the study, it is derived from the constitutional change that occurred in Brazil in 1997, which allowed mayors to run for one consecutive re-election, so a mayor can serve at most two consecutive terms (Ferraz and Finan 2011). Therefore, this would create a natural incentive for first-term mayors, as they would, presumably, seek re-election; in contrast, second-term mayors would have no such incentive. As a direct result of this difference in incentives, there is a difference in repercussions for engaging in corruption between first-term and second-term mayors because of the electorate's ability to exert their power more directly on a first-term mayor. To extrapolate this idea further, someone who desires to serve a second term as mayor must appease their electorate; therefore, this could lead to a conscious decision to minimize corruption, which illustrates a clear link for a potential causal relationship between re-election incentives and corruption. Moreover, it is essential to discuss the dataset, which represents a series of audits conducted by the Brazilian government, beginning in 2003, on municipalities' use of federal funds during the 2001–2004 mayoral term (Ferraz and Finan 2011). Additionally, the data was organized in such a way that it was observable if a mayor was in their first or second term.

Treatment and Identification Strategy:

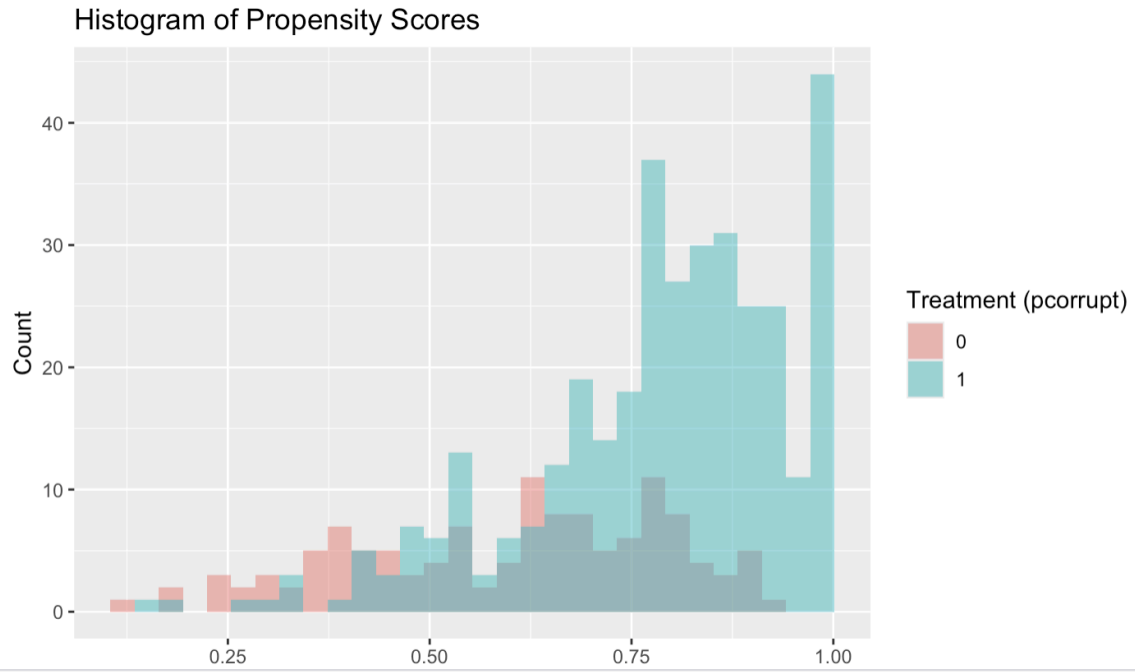
As mentioned above, the objective is to study the effect of election incentives on the likelihood of a mayor engaging in corruption; thus, it is necessary to define the treatment as the variable that determines if a mayor has re-election incentives, and this difference in incentives creates a clear treatment, which is directly related to whether or not the mayor is in their first or second term. Therefore, to state it explicitly, the treatment in this data is whether or not the mayor is in their first term, as I can assume that a mayor being in their first term has re-election incentives that allow them to be held more accountable to the public and minimize corruption, while a mayor in their second term has no such constraint due to their lack of re-election incentive. Hence, the treatment, D, is a binary indicator where:

- D = 1 if the mayor is serving their first term, which implies re-election incentives and increased electoral accountability.

- D = 0 if the mayor is serving their second term, which implies no re-election incentives and less electoral accountability.

Now, due to this data being measured over a series of different audits, where all of the chosen municipalities were selected randomly, I can assume that there is randomness in how each municipality was selected. However, this is not a randomized control trial. First and foremost, while the audits were random, the treatment, whether a mayor was in their first or second term, was not randomized because, in simple terms, mayors were not randomly assigned to be either in their first or second term. Instead, a mayor's term status is heavily dictated by observable and unobservable factors such as the mayor themselves, their district, the political landscape, and the electorate can influence whether a mayor is re-elected. Therefore, due to the fact that this data is not a result of a randomized trial, I can conclude that it comes from an observational study.

In order to derive causality from an observational study, I must assume unconfoundedness and overlap conditions hold within the dataset. Regarding unconfoundedness, it is mathematically represented by $Y(1), Y(0) \perp D \mid X$, which means that, conditional on a set of observed covariates X, the treatment assignment D is independent of the potential outcomes Y(1) and Y(0), and that will allow me to establish causality through a variety of machine learning methods. Furthermore, I must assume unconfoundedness because it cannot be proven in an observational data set. Moreover, the second important condition of an observational dataset is overlap, which definitionally means that, for every combination of covariates X, there is a positive probability of receiving both treatment (D = 1) and control (D = 0). By analyzing the propensity scores, I can assert that the overlap condition holds, as demonstrated by the select images below. With this treatment, identification strategy, and careful selection of covariates, I can isolate the effect of re-election incentives on corruption levels.


Histogram of Propensity Scores
Count
40
30
20
10
0
0.25
0.50
0.75
1.00
Treatment (pcorrupt)
0
1


Histogram of Propensity Scores
Count
30
20
10
0
0.70
0.75
0.80
0.85
Propensity Score
Treatment (ncorrupt_os)
0
1



Robustness of Covariates:

In order for my regression to be as accurate as possible, I must include all the covariates that I believe could potentially be confounders within the data. That is, they both influence the willingness of a mayor to engage in corruption, and they also affect the chance of a mayor being in their first or second term. In other words, the included covariates address potential confounding by controlling for factors that influence both the treatment variable (D, term status) and the outcome variable (Y, corruption).

For the first relationship, which is a covariate's influence on the level of corruption, I can assume such a relationship from economic intuition based on how certain covariates could change an individual's direct incentives for engaging in corruption. For the second relation, which is a covariate's influence on whether a mayor is a first or second term mayor, I can argue this through how that covariate influences a mayor's chance of getting elected. If a certain covariate increases a mayor's chance of being elected for a second term, then the probability of being a second term mayor when that covariate is high - or if it's present for binary covariates - would be high. For example, if I observe a much higher ratio of second term to first term mayors when comparing male mayors to female mayors, it would mean that the gender of a mayor affects the probability of re-election, and thus, it will have an effect on what I expect D to be.

Overall, I chose covariates that I believed had an effect on both the personal incentives a mayor would have to engage in corruption and the chances the mayor had of being re-elected. Thus, by including covariates that affect both corruption (Y) and term status (D), I aim to control

for confounding factors, thereby satisfying the unconfoundedness assumption and improving the validity of my causal estimates.

Summary of What I Chose:

- Mayoral Characteristics:
  - I selected Age and Education level as my confounders for this set. The reason being younger first-term mayors have longer career horizons, and therefore, they would be less incentivized to engage in acts of corruption versus older first-term mayors. For education, I believe that more educated first-term mayors could potentially have more of an ability to benefit from corruption without detection.
  - I believe that voters will view individuals who are older and more educated as being more experienced, and as a result, they will view them more favorably when considering to re-elect them for a second term compared to younger or less educated mayors.

- Municipal Characteristics:
  - I chose municipality characteristics based on if I believed they would increase the chances of a mayor being exposed for corruption and how a mayor being exposed for corruption would affect their chances of being re-elected.
  - I believed that some characteristics of a municipality may make elections more competitive, and as a result, it would reduce the chance of a mayor securing a second term if they engaged in unpopular acts such as corruption.

- Political and Judicial Characteristics:
  - I believe that the political climate could influence how a first or second term mayor engages in corruption.
  - A municipality with more competitive elections would not only reduce the chances of a mayor getting a second term, but it also reduces the likelihood of a first-term mayor engaging in corruption, as the increased competitiveness of elections would make the consequences of being exposed for being corrupt much greater and make their chances of being re-elected very small.

- Other:
  - I selected the state dummy as well because different states may have different geopolitical influences that could change the probability of being re-elected, while different state characteristics may also incentivize or disincentivize corruption.

After careful consideration, these are the covariates that I chose:

- **Mayoral Characteristics:**
  - Age
  - Schooling Level

- **Municipal Characteristics:**
  - Percentage of population that has at least a secondary education
  - Percentage of the population that lives in an urban sector
  - New municipality indicator
  - Log of GDP per capita in 2002
  - The Gini coefficient
  - The use of participatory budgeting in 2001 - 2004 administration

- **Political and Judicial Characteristics:**
  - The effective number of political parties in a legislature
  - The share of the legislature that is of the same party as the mayor

- **Others:**
  - State dummies

I have listed below each covariate that I considered, if I chose to include it or not, and my reasoning for its inclusion or exclusion. Ultimately, I chose to omit any covariate that I believed either had no effect on the level of corruption committed or no effect on the mayor's chance of re-election.

Reasoning behind Including/Not Including Covariates:

Age, Included: The older a mayor is, the less likely they are to fear imprisonment, given that they will not be able to fully live out their sentence. As a result, they may be more willing to engage in corruption. Other forms of long-term consequences for engaging in corruption, such as damage to reputation, are less damaging to someone who is older and further into their career. In conjunction, the reasons above demonstrate that age has an effect on the outcome. Moreover, old mayors presumably would have had longer careers, and as a result, they may be in their second term, or if not, they may be perceived as more experienced by the public, which increases their likelihood of re-election and reduces the incentive to avoid corruption in their first term.

Gender, Not Included: The incentives behind engaging in corruption would not change based on the mayor's gender, and the risk involved with corruption would not change either. Therefore, it

does not affect the outcome, and it also would not have an effect on the treatment due to a lack of systematic evidence between term status and gender. As a result, gender is not a confounder.

Schooling level, Included: Individuals who have a higher level of schooling may be less willing to engage in corruption, as they may be more aware of the long term consequences of choosing to engage in corruption. Alternatively, they could have more of an ability to benefit from corruption without detection, increasing their incentive to participate in corruption. Like with age, schooling level influences first-term mayors by increasing their re-election probability, and highly educated mayors could also be disproportionately serving in their second term.

Party Affiliation, Not Included: While factors such as how much of the legislature is of the same party could have an effect on corruption, the choice of party itself should not have an effect on the individual's choice to engage in corruption. Since Brazil has a large number of political parties, they cannot engage in more corruption than their competitors if they wish to stay relevant, so I do not expect members of some parties to be more corrupt than the members of others.

Log of Population, Not Included: I believe that the characteristics of the population, as well as how many resources the municipality receives is more important than the population of the municipality, thus I believe that Log of Population will not have an effect on the amount of corruption by a mayor.

Percentage of population with at least secondary education, Included: A more educated population will be more likely to be knowledgeable and invested in political affairs, and as a result, they would be much more involved in learning about corruption, making the risk and consequence of corruption much higher, disincentivizing corruption. Additionally, a more educated population would be more open to political progress, which would make their district more competitive and increase the number of first-term mayors.

Percentage of population in urban sector, Included: A higher percentage of the population being in an urban sector can indicate that a municipality has a larger economy; thus, there would be more funds circulating, and as a result, it would create a greater ease of using funds for corrupt purposes. Also, large urban cities can be extremely partisan, which would allow for more second-term mayors and a lack of repercussions for first-term mayors.

New municipality indicator, Included: A new municipality will most likely be much more chaotic, as the government is still trying to figure out how to operate effectively. Thus, it will be easier to siphon funds for corrupt purposes under the guise of waste from a newly established and not fully efficient section of the government. New districts may also still be searching for a political identity; therefore, they may have an increased number of first-term mayors.

Log of GDP per capita in 2002, Included: A higher GDP is an indicator of a larger and more active economy. I can usually expect a larger economy to have in turn a larger government with more public funds being sent to a wider array of locations. Given this increase in size, I can expect using the same amount of funds for corrupt purposes in a larger government is easier than in a smaller government, linking a higher GDP with an increased ease in engaging in corruption. Also, both wealthy and impoverished municipalities can be quite partisan, especially if they have been in existence for a while. Thus, they have more second-term mayors and a lack of repercussions for first-term mayors.

Gini Coefficient, Included: A higher Gini coefficient indicates more inequality within a municipality. Therefore, inequality may increase opportunities for corruption by reducing public accountability, as disadvantaged populations often lack the political power to hold officials accountable. Additionally, these environments favor entrenched elites or established political parties, making it more likely that mayors are serving in their second term and lessening the repercussions for first-term mayors.

Use of participatory budgeting in 2001-2004, Included: Citizens that engage publicly in the budgeting of a municipality will be much more invested in the allocation of funds. As a direct result, they will be more diligent and will have definitive proof if funds are being used for corrupt purposes. Thus, I expect that the use of participatory budgeting will make the risk of punishment for corruption much higher, incentivizing mayors to not engage in corruption. Furthermore, a more engaged population would be more open to political change, which would make their district more competitive and increase the number of first-term mayors.

Effective number of political parties in legislature, Included: By having more political parties in a legislature, it can be reasonably assumed that the municipality is more competitive in terms of elections. As a result, if a mayor is exposed for being corrupt, there will be more party options for the electorate, and additionally, there would be a wider body of opposing party members that would benefit from their opponents being exposed for corruption. As a result, the chances of not being re-elected if exposed for corruption in a municipality with more political parties is higher than if the municipality has a less competitive legislature. This means that the risk involved in corruption is higher, acting as a deterrent against corruption for first term mayors. Moreover, more competitive districts with a larger number of political parties would have more first-terms, as the electorate has numerous options if dissatisfied.

Number of legislators divided by number of voters, Not Included: Having more legislators per voter really only serves to indicate the size of the elected body of the government compared to the size of the municipality, therefore, it does not affect the outcome or the treatment.

Share of legislature that is of the same party as the mayor, Included: A higher share of the legislature being of the same party can indicate a less competitive government and less competitive elections. For similar reasons as to why I included the effective number of political parties in legislature, I believe that a higher share of legislature being of the same party makes elections less competitive, and thus, decreases the costs of being exposed for corruption, which incentivizes mayors to engage in corruption. Also, in municipalities dominated by a particular political party, there will be more second-term mayors and a lack of repercussions for first-term mayors.

Municipality being a judiciary district, Not Included: Assuming that a judiciary will preside over cases not just in their own district but also neighboring municipalities that are not judiciary districts, I assume that judiciaries will not be more or less aggressive in prosecuting the government of their local district compared to neighboring ones, as their oath would obligate them to faithfully preside over all mayors under their jurisdiction. As a result, a mayor being in a judiciary district or not being in a judiciary district would not change corruption incentives.

State Dummies, Included: Different states may be governed by different rules, and as a result the ability to commit corruption in a municipality in one state is higher than a municipality in a different state. It is important to include state dummy variables to create distinctions of corruption levels based on states as well as whether different states have a disproportionate number of mayors in their first or second term.

Lottery Dummies, Not Included: If the lottery picked municipalities at random, then what lottery you were part of should be random, and as a result, I should not expect any differences in the municipalities from one lottery or another. Thus, I believe that Lottery dummies will have no effect on the conditions of municipalities or corruption incentives and this is not a significant variable.

Data Results:

Pcorrupt Heterogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.019312425	0.009593319	-2.0131119	0.04466652	0.1016692	0.4952139
tree	tree	-0.022156392	0.009944282	-2.2280535	0.02639227	0.1107829	0.5365194
rf	rf	-0.020724470	0.009628008	-2.1525188	0.03185955	0.1055011	0.4969732
nnet	nnet	-0.004031784	0.010092109	-0.3994987	0.68972459	0.1243600	0.5371105

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.01802666	0.009470205	-1.903513	0.05757781	0.1015537	0.4969768
tree	tree	-0.01916793	0.009322748	-2.056039	0.04032934	0.1043805	0.5117342
rf	rf	-0.01684856	0.009927104	-1.697228	0.09030863	0.1061074	0.5138339
nnet	nnet	-0.01588734	0.010074830	-1.576934	0.11547925	0.1039312	0.5102624

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.02444051	0.01039011	-2.352286	0.01906560	0.1017668	0.4798373
tree	tree	-0.01913852	0.01041559	-1.837488	0.06680747	0.1151248	0.5462481
rf	rf	-0.02287272	0.01030529	-2.219513	0.02692341	0.1070520	0.4871167
nnet	nnet	-0.02020001	0.01211092	-1.667917	0.09606189	0.1322999	0.5096871

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.02196865	0.01002606	-2.191155	0.02892556	0.1000575	0.4798302
tree	tree	-0.01165134	0.01146192	-1.016526	0.30993377	0.1162100	0.5310962
rf	rf	-0.02444072	0.01016073	-2.405409	0.01653571	0.1048253	0.4877684
nnet	nnet	-0.01710257	0.01284057	-1.331917	0.18379304	0.1477089	0.5802094

4 rows

Pcorrupt Homogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.0193124252	0.0095933195	-2.013112	0.04466652	0.1016692	0.4952139
tree	tree	-0.0225769774	0.0092514139	-2.440381	0.01503625	0.1121053	0.5544937
rf	rf	-0.0207244696	0.0096280084	-2.152519	0.03185955	0.1055011	0.4969732
nnet	nnet	-0.0009601617	0.0001056103	-9.091557	0.00000000	0.1219736	7.1803297

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.01802666	0.009470205	-1.903513	0.05757781	0.1015537	0.4969768
tree	tree	-0.01935279	0.009040459	-2.140686	0.03280784	0.1039951	0.5177593
rf	rf	-0.01684856	0.009927104	-1.697228	0.09030863	0.1061074	0.5138339
nnet	nnet	-0.01583619	0.009991194	-1.585015	0.11362846	0.1037432	0.5113239

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.024440514	0.010390110	-2.3522864	0.01906560	0.1017668	0.4798373
tree	tree	-0.015670702	0.009807597	-1.5978126	0.11074966	0.1142921	0.5490324
rf	rf	-0.022872720	0.010305289	-2.2195128	0.02692341	0.1070520	0.4871167
nnet	nnet	0.003994557	0.027641402	0.1445135	0.88515625	4.4768160	0.5267963

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.02196865	0.010026059	-2.1911554	0.02892556	0.1000575	0.4798302
tree	tree	-0.01043307	0.010563732	-0.9876308	0.32383629	0.1147950	0.5369460
rf	rf	-0.02444072	0.010160732	-2.4054094	0.01653571	0.1048253	0.4877684
nnet	nnet	-0.02321141	0.009383651	-2.4736012	0.01372450	0.1479600	0.7138613

4 rows

Ncorrupt_os Heterogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.0045673479	0.004729136	-0.9657892	0.3346411	0.05035658	0.4947954
tree	tree	0.0007712885	0.004575850	0.1685563	0.8662224	0.05602087	0.5538881
rf	rf	-0.0033884997	0.004786432	-0.7079386	0.4793306	0.05125094	0.4962388
nnet	nnet	-0.0068483248	0.004955974	-1.3818322	0.1677374	0.05380952	0.5421358

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.007591117	0.004778615	-1.588560	0.1128251	0.05201823	0.4969758
tree	tree	-0.007709536	0.004773109	-1.615202	0.1069352	0.05268478	0.4968187
rf	rf	-0.005779416	0.004645715	-1.244032	0.2141015	0.05375063	0.5215555
nnet	nnet	-0.004693630	0.004427389	-1.060135	0.2896242	0.05223948	0.5135026

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.006549733	0.004693227	-1.395571	0.16349564	0.05040689	0.4801591
tree	tree	-0.005621382	0.005126567	-1.096520	0.27342516	0.05715887	0.5283844
rf	rf	-0.005374212	0.004649860	-1.155779	0.24835243	0.05092897	0.4866647
nnet	nnet	-0.008593002	0.004907803	-1.750886	0.08072268	0.05863693	0.5401827

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.007683101	0.004470309	-1.7186959	0.08632133	0.04691243	0.4801106
tree	tree	-0.004806096	0.004755276	-1.0106869	0.31269078	0.05230259	0.5417524
rf	rf	-0.005896759	0.004363266	-1.3514553	0.17719288	0.04829517	0.4881341
nnet	nnet	-0.002023487	0.005907683	-0.3425178	0.73216804	0.06734679	0.5786433

4 rows

Ncorrupt_os Homogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.007591117	0.004778615	-1.588560	0.11282514	0.05201823	0.4969758
tree	tree	-0.008175579	0.004797740	-1.704048	0.08902615	0.05292357	0.4994746
rf	rf	-0.005779416	0.004645715	-1.244032	0.21410149	0.05375063	0.5215555
nnet	nnet	-0.006423972	0.004530035	-1.418085	0.15682176	0.05264531	0.5167646

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.0045673479	0.004729136	-0.96578915	0.3346411	0.05035658	0.4947954
tree	tree	0.0001453513	0.004336771	0.03351602	0.9732772	0.05590659	0.5585892
rf	rf	-0.0033884997	0.004786432	-0.70793860	0.4793306	0.05125094	0.4962388
nnet	nnet	-0.0052232794	0.004317262	-1.20985932	0.2269346	0.05433091	0.5904288

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-0.006549733	0.004693227	-1.395571	0.16349564	0.05040689	0.4801591
tree	tree	-0.005155881	0.004911761	-1.049701	0.29438935	0.05642878	0.5319747
rf	rf	-0.005374212	0.004649860	-1.155779	0.24835243	0.05092897	0.4866647
nnet	nnet	-0.009167367	0.004085030	-2.244137	0.02528333	0.05662696	0.5705923

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-7.683101e-03	0.004470309	-1.71869588	0.08632133	0.04691243	0.4801106
tree	tree	-4.300970e-03	0.004630394	-0.92885635	0.35343531	0.05194330	0.5443801
rf	rf	-5.896759e-03	0.004363266	-1.35145535	0.17719288	0.04829517	0.4881341
nnet	nnet	-8.858156e-05	0.004509070	-0.01964519	0.98433466	0.06520568	0.6429075

4 rows

Valor_corrupt Heterogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-88162.42	54869.98	-1.606751	0.10877345	572261.5	0.4964112
tree	tree	-123936.05	60776.24	-2.039219	0.04198117	599742.7	0.5072806
rf	rf	-109297.94	58309.48	-1.874446	0.06148254	601080.7	0.5138196
nnet	nnet	-86853.95	56114.81	-1.547790	0.12234018	572114.3	0.5095651

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-116108.83	56745.28	-2.046141	0.04129259	565069.5	0.4925269
tree	tree	-91480.01	60816.59	-1.504195	0.13326018	652252.1	0.5486188
rf	rf	-106666.89	56456.84	-1.889353	0.05945335	581212.3	0.5019614
nnet	nnet	-113730.21	61749.60	-1.841797	0.06618389	577364.5	0.5258377

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-155268.72	63869.91	-2.431015	0.01542636	562274.2	0.4818717
tree	tree	-108773.62	69125.99	-1.573556	0.11631712	617753.0	0.5547995
rf	rf	-145785.92	64510.38	-2.259883	0.02428066	581713.6	0.4932275
nnet	nnet	-82883.09	64450.97	-1.285987	0.19917709	557491.2	0.5228724

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-156303.9	61971.18	-2.522204	0.01199005	552315.4	0.4825547
tree	tree	-129797.6	70897.63	-1.830775	0.06784250	606787.2	0.5501958
rf	rf	-152153.9	61240.35	-2.484537	0.01331543	570810.5	0.4944573
nnet	nnet	-114509.2	51863.03	-2.207916	0.02790377	561831.8	0.5557654

4 rows

Valor_corrupt Homogeneous for 1) Mayoral, 2) Mayoral and Municipal, 3) Mayoral, Municipal, and Political, and 4) All Selected Covariates):

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-88162.42	54869.98	-1.606751	0.10877345	572261.5	0.4964112
tree	tree	-123936.05	60776.24	-2.039219	0.04198117	599742.7	0.5072806
rf	rf	-109297.94	58309.48	-1.874446	0.06148254	601080.7	0.5138196
nnet	nnet	-86792.47	56114.06	-1.546715	0.12259787	571570.6	0.5090332

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-116108.83	56745.28	-2.046141	0.04129259	565069.5	0.4925269
tree	tree	-71899.57	55594.11	-1.293295	0.19653770	641395.2	0.5587847
rf	rf	-106666.89	56456.84	-1.889353	0.05945335	581212.3	0.5019614
nnet	nnet	-111279.11	52272.91	-2.128810	0.03378384	569390.5	0.5573190

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-155901.1	63288.06	-2.463358	0.01411774	561779.1	0.4835799
tree	tree	-110651.2	61406.27	-1.801953	0.07218668	618121.9	0.5691691
rf	rf	-145785.9	64510.38	-2.259883	0.02428066	581713.6	0.4932275
nnet	nnet	-128167.6	58225.59	-2.201224	0.02820040	562751.5	0.5513914

4 rows

	model <chr>	theta <dbl>	std_error <dbl>	t_value <dbl>	p_value <dbl>	rmse_g <dbl>	rmse_m <dbl>
lasso	lasso	-150447.59	60022.98	-2.506500	0.01252638	550727.6	0.4895868
tree	tree	-112714.31	59307.10	-1.900520	0.05797010	598088.7	0.5714554
rf	rf	-152153.91	61240.35	-2.484537	0.01331543	570810.5	0.4944573
nnet	nnet	-80531.55	38337.98	-2.100568	0.03620543	559010.4	0.6807962

4 rows

Robustness of Various Control Variables:

To analyze the model's robustness under different control variables, I examined the causal question under my machine learning framework while methodically adding covariates that I believed to be confounders. I segmented the additions in categories as follows:

1. **Mayoral Characteristics**: Age, education level.
2. **Mayoral and Municipal Characteristics**: Adds urban sector share, log GDP per capita, secondary education share, new municipality indicator, Gini coefficient, and the use of participatory budgeting in 2001 - 2004 administration to the previous covariates.
3. **Mayoral, Municipal, and Political Characteristics**: Adds the effective number of political parties and the share of the legislature aligned with the mayor to the previous covariates.
4. **All Covariates**: Includes state dummies to account for state-level heterogeneity to the previous covariates.

By inspection, it is immediately apparent that, under all statistically significant results, p-value < .05, and almost all of the other results, theta, the treatment effect, is negative, which does indicate that re-election incentives do reduce corruption. Furthermore, I see that, in general, theta becomes more negative as more of the selected covariates are added. This increase in magnitude represents the salience of the selected covariates, providing a form of verification that they are confounders. For example, if I look at the theta of the lasso machine learning method for the measure of corruption, valor_corrupt, I see that the analysis on the model that only accounts for mayoral characteristics and the model that incorporates all selected covariates yields two completely different numbers, which are the following: the Lasso estimate grows from theta = -88,162.42 (Mayoral only) to theta = - 150,447.59 (All Covariates). In summation, I see that, under all different examined permutations of how I segmented my data, my analysis confirms that omitted variable bias does not undermine the observed causal relationship, and the treatment is robust to the inclusion of different control variables.

Robustness of Different Machine Learning Methods:

For this study, I used four different machine learning methods to discern the causality of election incentives on corruption level: A Lasso regression, a Random Forest regression, a Trees regression, and a Single Layer Neural Network regression. By using all four methods, I am able to determine which method is most accurate by finding the method with the smallest root mean squared error (RMSE). Testing multiple methods ensures the results are not sensitive to the assumptions or limitations of any specific model.

To analyze the robustness across machine learning models, I attempted to determine the treatment effect for the three different corruption measures using the following techniques:

1. **Lasso Regression**: Regularizes the coefficients to prevent overfitting.
2. **Tree Regression**: Captures non-linear relationships but overfitting is possible.
3. **Random Forest (RF)**: An ensemble method that averages multiple decision trees to improve stability and accuracy.
4. **Neural Network (NNET)**: Captures complex interactions; however, it is prone to overfitting or exhibiting instability with small sample sizes.

As with checking for robustness to the inclusion of different control variables, I can again assert, by inspection of the data tables above, that the treatment effect, theta, is consistently negative, which confirms the robust nature of the causal estimate across different machine learning techniques. With that said, I can see a difference in magnitude between the models. For instance, in valor_corrupt, Random Forest estimates theta = - 152,153.91, while Lasso estimates theta = - 150,447.59. These differences could plausibly be the result of different sensitivities to

how the model handles calculations during the analysis of the treatment effect. I can also see that there is a significant amount of similarity in the RMSEs and the p-values of the estimates across different machine learning models. This, in conjunction with the estimates having the same directionality (all being negative) and relatively similar magnitudes, allows me to conclude that the causal estimates are robust to the use of different machine learning techniques.

Robustness of Homogeneous or Heterogeneous Treatment Model:

I ran each machine learning method on both a homogeneous treatment effect model and a heterogeneous treatment effect model, using my custom DoubleMLPLR and DoubleMLIRM frameworks. To elaborate on my methodology, I use the PLR (Partially Linear Regression) to simulate a homogeneous treatment effect because it assumes a single, constant treatment effect across all observations, and I used the IRM (Interactive Regression Model) to simulate a heterogeneous treatment model because it allows the treatment effect to vary with covariates, estimating subgroup-specific treatment effects (Chernozhukov et al. 2018). This approach ensures the robustness of my estimations by accounting for potential variability in the true underlying data-generating process.

Again, as with the above, the treatment effect, theta, is consistently negative, which confirms the robust nature of the causal estimate under both the homogeneous and heterogeneous treatment effect model. For example, while the effects of the heterogeneous model are, in general, larger in magnitude, they are still both fairly similar, and the difference could plausibly be explained by how the two models account for the effect of the treatment. Specifically, I can look at the Lasso of the corruption measure, valor_corrupt, to demonstrate this slight difference, as the theta for the heterogeneous model is - 156,303.9, while the theta for the homogeneous model is - 150,447.59. Moreover, like with the above, I see similar RMSE and other statistical measures, which further points to the causal estimate's robustness to heterogeneous and homogeneous treatment effect models.

Robustness of Different Measures of Corruption:

There are three separate measures of corruption within this study: the amount of corruption, i.e. the value of audited resources involved in corruption (valor_corrupt), the share of audited resources found involved in corruption (pcorrupt), and the number of corruption-related irregularities per audited service item (ncorrupt_os). With regressions on three different definitions, I am able to quantify corruption and examine the treatment effect from a comprehensive view. Given that corruption may take different forms in the real world, I undoubtedly enhance the robustness of my estimation by taking into account the different measures of corruption, as robustness to these alternative definitions ensures that the findings are not an artifact of a specific choice of metric.

Again, as with the above, the treatment effect, theta, is consistently negative, which confirms that re-election incentives do reduce corruption across the different measures of corruption. Moreover, the consistency of theta and the other statistical measures across the different definitions of corruption implies that the causal estimate is robust across different measures of corruption. Overall, this further analysis of robustness helps affirm the reliability of the model to derive meaningful conclusions from the data across various measure operationalizations.

References:

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